WORKED SOLUTION

STEP III 1989, Question 2

Pure • Planes in Three Dimensions

STEP III 1989, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q2

The key idea

Put in coordinates: east, north and up. Then each plane is a graph $z = px + qy + r$, and three points fix each one. How steep a plane is depends only on how fast z changes horizontally, so the angle comes from p and q. For the second part there is a neat shortcut: the *depth* of the seam below the ground is also a linear function of position, and the seam reaches the surface exactly where that depth is zero.

Solution

Setting up

Take A as the origin, with the x-axis pointing east, the y-axis north and the z-axis vertically up (all lengths in metres). Then

$$A = (0,0,0), \qquad B = (0,100,-5), \qquad C = (60,100,-10),$$

since B is 5 m below A and C is a further 5 m below B. The bore-holes are vertical, so the points where they strike the seam are directly below these:

$$A' = (0,0,-95), \qquad B' = (0,100,-50), \qquad C' = (60,100,-86).$$

The angle of the seam

The seam is not vertical, since A', B', C' lie above three points that are not collinear in plan. So we can write it as $z = px + qy + r$.

- At A': $r = -95$.
- From A' to B', y increases by 100 and z by 45, so $100q = 45$ and $q = \frac{9}{20}$.
- From B' to C', x increases by 60 and z changes by -36, so $60p = -36$ and $p = -\frac{3}{5}$.

So the seam is

$$z = -\tfrac{3}{5}x + \tfrac{9}{20}y - 95,$$

and a normal vector to it is $\mathbf{n} = \left(\frac{3}{5}, -\frac{9}{20}, 1\right)$, or, scaling by 20, $(12, -9, 20)$.

The angle between two planes equals the angle between their normals. The horizontal plane has normal $\mathbf{k} = (0, 0, 1)$, so the angle θ of the seam to the horizontal satisfies

$$\cos\theta = \frac{|(12,-9,20)\cdot(0,0,1)|}{\sqrt{12^2+9^2+20^2}} = \frac{20}{\sqrt{144+81+400}} = \frac{20}{\sqrt{625}} = \frac{20}{25} = \frac{4}{5}.$$

Hence the seam is inclined at $\cos^{-1}\left(\frac{4}{5}\right)$ to the horizontal, as required.

(Another way to see it: along the line of greatest slope, z changes by $\sqrt{p^2+q^2} = \sqrt{\frac{9}{25}+\frac{81}{400}} = \frac{3}{4}$ per metre horizontally. So $\tan\theta = \frac{3}{4}$, a 3–4–5 triangle, and $\cos\theta = \frac{4}{5}$.)

Where the seam reaches the surface

The ground is a plane through the origin. In the same way as before, going north from A to B drops it by 5 in 100, and going east from B to C drops it by 5 in 60, so

$$z = -\tfrac{1}{12}x - \tfrac{1}{20}y.$$

Ask yourself Do we really need to solve two plane equations at the same time?

Not quite. Let $d(x, y)$ be the depth of the seam below the ground at the point with horizontal position (x, y). It is the difference of two linear functions, so it is linear itself:

$$d(x,y) = \left(-\tfrac{1}{12}x - \tfrac{1}{20}y\right) - \left(-\tfrac{3}{5}x + \tfrac{9}{20}y - 95\right) = \tfrac{31}{60}x - \tfrac{1}{2}y + 95.$$

As a check, $d(0,0) = 95$, $d(0,100) = 45$ and $d(60,100) = 45 + 31 = 76$, which are the three bore-hole depths. (Indeed we could have written d down straight from these: the depth falls by 50 in 100 m going north and rises by 31 in 60 m going east.)

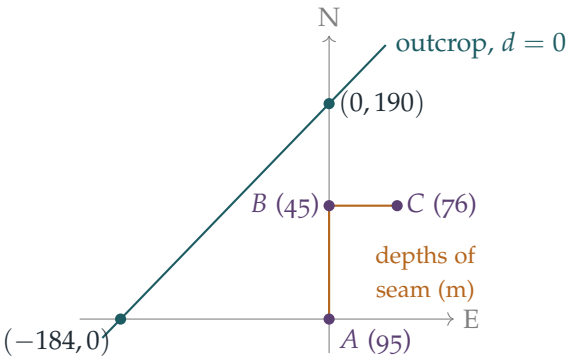
The seam comes to the surface where $d = 0$, which in plan is the line

$$\tfrac{31}{60}x - \tfrac{1}{2}y + 95 = 0, \qquad \text{that is,} \qquad y = \tfrac{31}{30}x + 190.$$

Along this line, moving 30 m east goes with moving 31 m north. So the line makes an angle φ with due north, where

$$\tan\varphi = \frac{\text{east}}{\text{north}} = \frac{30}{31}, \qquad \varphi = \tan^{-1}\left(\tfrac{30}{31}\right) \approx 44.06^\circ.$$

The bearing of the line is therefore $\tan^{-1}\left(\frac{30}{31}\right) \approx 044^\circ$. Read in the opposite direction it is 224°; a line has both bearings, and either describes it. The line runs just west of north-east, and in plan it passes 190 m due north of A.



Checks

- The seam's normal $(12, -9, 20)$ has length exactly 25, so $\cos\theta = \frac{20}{25}$, a clean 3–4–5 answer, as the question promised.
- The depth formula d reproduces all three bore-hole depths, so both planes are right.
- Numerically, $\tan^{-1}(30/31) = 44.0608°$. Random points on the line $y = \frac{31}{30}x + 190$ give the same height on both planes, to rounding error.

Common slips

- Using depths below *each* hole's top as if the tops were level. B and C are 5 m and 10 m lower than A, so the seam is 50 m and 86 m below A's level there, not 45 m and 76 m.
- Giving the bearing of a horizontal line in the seam (where $-\frac{3}{5}x + \frac{9}{20}y$ is constant, bearing $\tan^{-1}\frac{3}{4} \approx 037°$) instead of the outcrop. Because the ground slopes too, these differ.
- Mixing up $\tan\varphi = \text{east}/\text{north}$ with north/east. Bearings are measured clockwise from north.

Beyond STEP

This is the geologist's problem of "strike and dip", and the tools are those of first-year vector geometry: planes as $\mathbf{n} \cdot \mathbf{r} = c$, angles between planes through their normals, and lines of intersection. The depth trick is an instance of a general principle: the difference of two affine functions is affine, so the intersection of two non-parallel planes is the zero set of one linear function. Strang's *Introduction to Linear Algebra* builds this picture into the geometry of linear equations.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.